



Qualification Pack

Reverse Engineering and Additive Manufacturing QA Supervisor

QP Code: CUTM/CSC/Q0101

Version: 1.0

NSQF Level: 5

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CUTM/CSC/Q0101: Reverse Engineering and Additive Manufacturing QA Supervisor

Brief Job Description

Reverse Engineering and Additive Manufacturing QA Supervisor, also called Rapid Prototype Specialist is responsible for understanding the design requirements, material, generate design program for additive manufacturing of various parts 3D Printer, assemble it and inspect the parts to finalize and deliver the work piece. Also maintain good condition of the machine with regular maintenance.

Personal Attributes

A Reverse Engineering and Additive Manufacturing QA Supervisor should possess strong analytical skills, attention to detail, problem-solving ability, and a keen understanding of quality standards. Effective communication, leadership, and teamwork are essential, along with adaptability to evolving technologies. A commitment to precision and continuous improvement is crucial in this role.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [CUTM/CSC/N0101: 3D digitizing, scan to CAD and interpretation of designs](#)
2. [CUTM/CSC/N0102: Programming, Metrology, Maintenance, and repair of 3D machine](#)
3. [CUTM/CSC/N0103: Operate 3D Printing machine](#)
4. [CUTM/CSC/N0104: Inspect, Finalize, and deliver the product](#)
5. [CUTM/VSQ/N7801: Ensure organizational health, safety, and security](#)
6. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Capital Goods
Sub-Sector	Advanced Machining
Occupation	Additive Manufacturing
Country	India
NSQF Level	5

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Credits	18
Aligned to NCO/ISCO/ISIC Code	NSQF
Minimum Educational Qualification & Experience	B.E./B.Tech (completed 3rd year in Mechanical, Production, Mechatronics.) with NA of experience OR B.E./B.Tech (completed 4th year in Mechanical, Production, Mechatronics.) with NA of experience OR Completed 2nd year diploma after 12th (in Mechanical or Production) with NA of experience OR Completed 3 year diploma after 10th (Mechanical or Production) with 1.5 years of experience in relevant experience OR Previous relevant Qualification of NSQF Level (4) with 3 Years of experience in relevant experience OR Previous relevant Qualification of NSQF Level (4.5) with 1.5 years of experience in relevant experience
Minimum Level of Education for Training in School	12th Class
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	31/01/2027
NSQC Approval Date	31/01/2024
Version	1.0
Reference code on NQR	QG-05-CG-01974-2024-V1-CUTM
NQR Version	1.0

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CUTM/CSC/N0101: 3D digitizing, scan to CAD and interpretation of designs

Description

This NOS focuses on 3D digitizing, converting scan data into CAD models, and interpreting designs for manufacturing and quality assurance. It involves operating 3D scanning equipment, refining scanned data, and ensuring accuracy in CAD models for reverse engineering applications while maintaining industry standards and quality control.

Scope

The scope covers the following :

- The scope includes handling 3D scanning technologies, processing scan data for CAD conversion, and verifying design accuracy. It applies to industries such as automotive, aerospace, and medical device manufacturing, where precise digital modeling, reverse engineering, and quality assurance are critical for product development and defect analysis.

Elements and Performance Criteria

3D digitizing

To be competent, the user/individual on the job must be able to:

- PC1.** perform equipment's adjustment and calibration process.
- PC2.** make technical decisions regarding the possibility of optical 3D digitizing Technical: Possible or impossible to perform, the accuracy it is possible to attain for the object, required conditions for digitizing.
- PC3.** make decisions regarding pre-process work, Pre-process work: disassembly, washing, painting
- PC4.** perform pre-process actions for applying matted coatings.
- PC5.** apply matted coatings and optical marks.
- PC6.** coordinate to fix objects for subsequent digitization.
- PC7.** perform 3D digitizing for various objects with different materials, surface characteristics, geometrical complexity.
- PC8.** ensure to save results in the required form.

Scan to CAD and optimize

To be competent, the user/individual on the job must be able to:

- PC9.** create editable CAD models by digitized data (polygonal models)
- PC10.** apply mathematical modeling of additive manufacturing technologies to improve the performance of device, machine, and mechanism parts.
- PC11.**
 - restore missing data of elements of objects to be redesigned, from available data:
 - • of polygonal models (for example, the gear wheel has only one preserved tooth, the worm has only one turn, or there is only one-third of the flange)
 - • taken from connected parts
 - • taken from existing objects by manual measuring (for example, the depth of a blind hole)

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- PC12.** change geometry of created models according to task.
- PC13.** consider features of AM and subsequent finishing processing.
- PC14.** analyze and optimize the structure of the model in accordance with the terms of reference.
- PC15.** analyze the deviation of created models from the results of 3D scanning.
- PC16.** provide lattices and surfaces' topology, with analysis and optimization according to task.
- PC17.** apply standards for conventional dimensioning and tolerances, and geometric dimensioning and tolerance appropriate to the ISO standard.

Interpretation of designs and drawings

To be competent, the user/individual on the job must be able to:

- PC18.** interpret technical drawings and specifications of the product.
- PC19.** identify critical features of the product, surface finish and geometric requirements.
- PC20.** analyze manufacturability with the available resources.
- PC21.** identify and prepare for potential assembly issues if any
- PC22.** identify and prepare for any maintenance issues that may arise during production.
- PC23.** specify stock to be kept for different operations.
- PC24.** plan production of parts according to specifications.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Fundamentals of 3D machine programming and automation
- KU2.** Basics of metrology and precision measurement techniques.
- KU3.** Calibration methods for ensuring measurement accuracy.
- KU4.** Common errors in 3D scanning and methods for correction.
- KU5.** Preventive maintenance techniques for 3D machines.
- KU6.** Troubleshooting mechanical and software issues in 3D machines.
- KU7.** Industry standards and regulatory compliance for 3D machine operation.
- KU8.** Applications of metrology in quality assurance and manufacturing.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** operate 3D scanners and related software
- GS2.** interpret and analyze scan data for CAD conversion.
- GS3.** use Problem-solving skills for error correction in 3D models.
- GS4.** use CAD tools for reverse engineering applications.
- GS5.** do quality assessment and validation of 3D-scanned models.
- GS6.** do effective communication and collaboration with engineering teams
- GS7.** Adapt to advancements in scanning and CAD technologies.
- GS8.** Decision-making skills for process optimization in scan-to-CAD workflows.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>3D digitizing</i>	10	9	-	15
PC1. perform equipment's adjustment and calibration process.	1	1	-	2
PC2. make technical decisions regarding the possibility of optical 3D digitizing Technical: Possible or impossible to perform, the accuracy it is possible to attain for the object, required conditions for digitizing.	1	1	-	3
PC3. make decisions regarding pre-process work, Pre-process work: disassembly, washing, painting	2	1	-	1
PC4. perform pre-process actions for applying matted coatings.	1	1	-	3
PC5. apply matted coatings and optical marks.	1	1	-	2
PC6. coordinate to fix objects for subsequent digitization.	1	1	-	1
PC7. perform 3D digitizing for various objects with different materials, surface characteristics, geometrical complexity.	2	1	-	2
PC8. ensure to save results in the required form.	1	2	-	1
<i>Scan to CAD and optimize</i>	11	15	-	11
PC9. create editable CAD models by digitized data (polygonal models)	1	1	-	2
PC10. apply mathematical modeling of additive manufacturing technologies to improve the performance of device, machine, and mechanism parts.	2	1	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. • restore missing data of elements of objects to be redesigned, from available data: • • of polygonal models (for example, the gear wheel has only one preserved tooth, the worm has only one turn, or there is only one-third of the flange) • • taken from connected parts • • taken from existing objects by manual measuring (for example, the depth of a blind hole)	1	1	-	2
PC12. change geometry of created models according to task.	1	1	-	3
PC13. consider features of AM and subsequent finishing processing.	1	1	-	2
PC14. analyze and optimize the structure of the model in accordance with the terms of reference.	1	1	-	1
PC15. analyze the deviation of created models from the results of 3D scanning.	2	3	-	-
PC16. provide lattices and surfaces' topology, with analysis and optimization according to task.	1	3	-	-
PC17. apply standards for conventional dimensioning and tolerances, and geometric dimensioning and tolerance appropriate to the ISO standard.	1	3	-	-
<i>Interpretation of designs and drawings</i>	9	16	-	4
PC18. interpret technical drawings and specifications of the product.	2	3	-	-
PC19. identify critical features of the product, surface finish and geometric requirements.	1	1	-	2
PC20. analyze manufacturability with the available resources.	1	1	-	2
PC21. identify and prepare for potential assembly issues if any	1	3	-	-
PC22. identify and prepare for any maintenance issues that may arise during production.	1	3	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC23. specify stock to be kept for different operations.	1	3	-	-
PC24. plan production of parts according to specifications.	2	2	-	-
NOS Total	30	40	-	30



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National Occupational Standards (NOS) Parameters

NOS Code	CUTM/CSC/N0101
NOS Name	3D digitizing, scan to CAD and interpretation of designs
Sector	Capital Goods
Sub-Sector	
Occupation	Additive Manufacturing
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2027
NSQC Clearance Date	31/01/2024

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CUTM/CSC/N0102: Programming, Metrology, Maintenance, and repair of 3D machine

Description

This NOS covers programming, metrology, maintenance, and repair of 3D machines used in reverse engineering and additive manufacturing. It involves precision measurement, machine calibration, troubleshooting, and preventive maintenance to ensure accuracy and efficiency in 3D scanning and printing, supporting quality assurance in manufacturing processes.

Scope

The scope covers the following :

- The scope includes programming 3D machines, implementing metrology techniques for accuracy, performing routine maintenance, and troubleshooting technical issues. It applies to industries such as aerospace, automotive, and healthcare, where high-precision 3D scanning and additive manufacturing are essential for quality control, component validation, and product development.

Elements and Performance Criteria

Create programming

To be competent, the user/individual on the job must be able to:

- PC1.** select the best sequence for production of parts based on type and part specification.
- PC2.** use skill specific software and related hardware.
- PC3.** generate programs using software.
- PC4.** transfer programs to machines.
- PC5.** create or edit programs directly on the control.

Study metrology requirements

To be competent, the user/individual on the job must be able to:

- PC6.** prepare objects and measuring instruments for measurements.
- PC7.** calibrate, adjust, and align measuring instruments.
- PC8.** select correct measuring instruments and devices (styluses, probes, etc.), auxiliary and fixing devices (vices, Vblocks, clamps, etc.), relating to measurement strategy.
- PC9.** perform measurements using various control and measuring instruments.
- PC10.** read the indications of measuring instruments.
- PC11.** identify and estimate the correctness of measurements and the reliability of the data obtained, to minimize associated human factor errors.
- PC12.** find the required information in specialized reference books, tables, or diagrams.
- PC13.** carry out routine maintenance of measuring instruments.

Maintenance and repair of machining equipment

To be competent, the user/individual on the job must be able to:

- PC14.** do preventative maintenance to avoid issues arising.
- PC15.** make repairs when problem arise.



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Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Fundamentals of 3D machine programming and automation.
- KU2.** Basics of metrology and precision measurement techniques.
- KU3.** Calibration methods for ensuring measurement accuracy.
- KU4.** Common errors in 3D scanning and methods for correction.
- KU5.** Preventive maintenance techniques for 3D machines.
- KU6.** Troubleshooting mechanical and software issues in 3D machines.
- KU7.** Applications of metrology in quality assurance and manufacturing.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** program 3D machines for automated tasks.
- GS2.** operate and calibrate metrology equipment.
- GS3.** Troubleshoot and problem-solving skills for machine errors.
- GS4.** perform maintenance and minor repairs.
- GS5.** do quality control and validation of scanned or printed parts.
- GS6.** report technical issues and solutions.
- GS7.** adhere safety and compliance standards in machine operation.
- GS8.** Adapt to new technologies and software updates in 3D metrology.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Create programming</i>	14	20	-	-
PC1. select the best sequence for production of parts based on type and part specification.	3	4	-	-
PC2. use skill specific software and related hardware.	3	4	-	-
PC3. generate programs using software.	3	4	-	-
PC4. transfer programs to machines.	2	4	-	-
PC5. create or edit programs directly on the control.	3	4	-	-
<i>Study metrology requirements</i>	20	32	-	-
PC6. prepare objects and measuring instruments for measurements.	2	4	-	-
PC7. calibrate, adjust, and align measuring instruments.	3	4	-	-
PC8. select correct measuring instruments and devices (styluses, probes, etc.), auxiliary and fixing devices (vices, Vblocks, clamps, etc.), relating to measurement strategy.	3	4	-	-
PC9. perform measurements using various control and measuring instruments.	3	4	-	-
PC10. read the indications of measuring instruments.	2	4	-	-
PC11. identify and estimate the correctness of measurements and the reliability of the data obtained, to minimize associated human factor errors.	3	4	-	-
PC12. find the required information in specialized reference books, tables, or diagrams.	2	4	-	-
PC13. carry out routine maintenance of measuring instruments.	2	4	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Maintenance and repair of machining equipment</i>	6	8	-	-
PC14. do preventative maintenance to avoid issues arising.	3	4	-	-
PC15. make repairs when problem arise.	3	4	-	-
NOS Total	40	60	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	CUTM/CSC/N0102
NOS Name	Programming, Metrology, Maintenance, and repair of 3D machine
Sector	Capital Goods
Sub-Sector	
Occupation	Additive Manufacturing
NSQF Level	5
Credits	4
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2027
NSQF Clearance Date	31/01/2024

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CUTM/CSC/N0103: Operate 3D Printing machine

Description

This NOS covers the operation of 3D printing machines, including machine setup, material selection, print optimization, and troubleshooting. It ensures adherence to industry standards and safety guidelines while producing high-quality printed parts for reverse engineering and additive manufacturing applications across various industries.

Scope

The scope covers the following :

- The scope includes operating, calibrating, and maintaining 3D printers, optimizing print parameters, and ensuring quality output. It applies to industries such as aerospace, healthcare, automotive, and consumer goods manufacturing, where precise 3D printing is essential for prototyping, production, and quality assurance in additive manufacturing.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Follow their selected process strategy.
- PC2.** Follow a given process- strategy (external Additive Manufacturing programme)
- PC3.** Upload generated programmes to 3D printing.
- PC4.** Mount and align base plate.
- PC5.** Identify and designate the different processes.
- PC6.** Optimize the process strategy.
- PC7.** Define and adjust the parameters as a function of the operation sequence, material type, type of operation
- PC8.** Start the process from the raw material.
- PC9.** Perform the Additive Manufacturing operations.
- PC10.** Avoid vibration in the machining sequences.
- PC11.** Record the machine parameter.
- PC12.** Optimize machining strategies.
- PC13.** Quickly react if anything goes wrong (making emergency stops)
- PC14.** Obtain dimensions, geometries, surface roughness.
- PC15.** Make the final parts conform to blueprints.
- PC16.** Report health, safety, and environmental issues to the appropriate personnel.
- PC17.** Report equipment failure to the appropriate personnel.
- PC18.** Note all the production related information on the respective register, log, work sheet.
- PC19.** Update all production information on the MIS system.
- PC20.** Store the approved program against the batch number for future reference.

Qualification Pack

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Fundamentals of additive manufacturing and 3D printing processes
- KU2.** Types of 3D printing technologies and their applications
- KU3.** Material properties and selection for 3D printing.
- KU4.** Machine setup, calibration, and operational procedures.
- KU5.** Importance of layer resolution, speed, and temperature control.
- KU6.** Common defects in 3D printing and troubleshooting methods.
- KU7.** Post-processing techniques such as support removal and surface finishing.
- KU8.** Industry standards and quality control measures in 3D printing.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** operate and calibrate different types of 3D printers.
- GS2.** slicing software for print preparation and optimization.
- GS3.** Troubleshoot and problem-solving skills for print defects.
- GS4.** do quality assessment and validation of printed parts.
- GS5.** report issues and collaborating with teams.
- GS6.** Adhere to safety and compliance standards in Additive Manufacturing processes.
- GS7.** Adapt to new materials and emerging 3D printing technologies.
- GS8.** use decision-making skills for selecting print parameters and optimizing output.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	40	60	-	-
PC1. Follow their selected process strategy.	2	3	-	-
PC2. Follow a given process- strategy (external Additive Manufacturing programme)	2	3	-	-
PC3. Upload generated programmes to 3D printing.	2	3	-	-
PC4. Mount and align base plate.	2	3	-	-
PC5. Identify and designate the different processes.	1	2	-	-
PC6. Optimize the process strategy.	2	3	-	-
PC7. Define and adjust the parameters as a function of the operation sequence, material type, type of operation	2	3	-	-
PC8. Start the process from the raw material.	1	2	-	-
PC9. Perform the Additive Manufacturing operations.	4	6	-	-
PC10. Avoid vibration in the machining sequences.	2	3	-	-
PC11. Record the machine parameter.	2	3	-	-
PC12. Optimize machining strategies.	2	3	-	-
PC13. Quickly react if anything goes wrong (making emergency stops)	2	3	-	-
PC14. Obtain dimensions, geometries, surface roughness.	2	3	-	-
PC15. Make the final parts conform to blueprints.	2	3	-	-
PC16. Report health, safety, and environmental issues to the appropriate personnel.	2	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC17. Report equipment failure to the appropriate personnel.	2	3	-	-
PC18. Note all the production related information on the respective register, log, work sheet.	2	3	-	-
PC19. Update all production information on the MIS system.	2	3	-	-
PC20. Store the approved program against the batch number for future reference.	2	2	-	-
NOS Total	40	60	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	CUTM/CSC/N0103
NOS Name	Operate 3D Printing machine
Sector	Capital Goods
Sub-Sector	
Occupation	Additive Manufacturing
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2027
NSQF Clearance Date	31/01/2024

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CUTM/CSC/N0104: Inspect, Finalize, and deliver the product

Description

This NOS focuses on inspecting, finalizing, and delivering products manufactured through reverse engineering and additive manufacturing. It ensures quality control, defect detection, post-processing, and compliance with industry standards. Professionals in this role validate product accuracy, functionality, and readiness before dispatching to clients or further production stages.

Scope

The scope covers the following :

- The scope includes product inspection using metrology techniques, quality validation, post-processing, packaging, and final delivery. It applies to industries like aerospace, automotive, healthcare, and consumer goods, where precision, compliance, and product integrity are critical for successful reverse engineering and additive manufacturing applications.

Elements and Performance Criteria

Inspect the manufactured products

To be competent, the user/individual on the job must be able to:

- PC1.** complete the post processing methods.
- PC2.** clean the parts for inspection.
- PC3.** check visual inspection on completion of all the process as per drawing.
- PC4.** check the calibration status of various measuring instrument.
- PC5.** identify parameter wise measuring instruments for measurement.
- PC6.** inspect the machined product with various measurement tools like vernier caliper, micrometer.
- PC7.** check the surface finish as per requirements.
- PC8.** check the fitness of the product using go or no-go devices.
- PC9.** check visual inspection on the damage, burr, excess materials.
- PC10.** document all the measured data for various parameters.

Finalize and deliver the product

To be competent, the user/individual on the job must be able to:

- PC11.** clean the parts.
- PC12.** pack the parts in proper way to avoid any damages during transport.
- PC13.** complete the delivery documents.
- PC14.** deliver the parts, drawings, and digital memory devices to the appropriate location and/or personnel as required by the organization.
- PC15.** dismount tools, clamping, accessories.
- PC16.** clean the machines.
- PC17.** set the environment to its initial state, ready for the next job.
- PC18.** evaluate and report on factors and outcomes relevant to requirements and expectations.

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Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Principles of quality control and inspection in reverse engineering and additive manufacturing
- KU2.** Industry standards and tolerances for final product validation
- KU3.** Techniques for visual, dimensional, and functional inspection of 3D-printed and reverse-engineered parts
- KU4.** Common defects in manufactured components and methods for rectification
- KU5.** Use of metrology tools such as calipers, micrometers, and CMM (Coordinate Measuring Machines)
- KU6.** Documentation and reporting procedures for quality assurance.
- KU7.** Post-processing techniques to enhance product quality (sanding, polishing, coating, etc.)
- KU8.** Packaging and handling requirements for safe product delivery

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** use measurement and inspection tools.
- GS2.** identify and rectify defects in manufactured products.
- GS3.** do documentation and quality reporting.
- GS4.** use problem-solving skills for finalizing product quality.
- GS5.** do effective communication with production teams and clients.
- GS6.** Adhere to safety and quality control regulations.
- GS7.** use decision-making skills for product approval and rejection criteria.
- GS8.** use the knowledge of logistics and handling for final product dispatch.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Inspect the manufactured products</i>	24	36	-	-
PC1. complete the post processing methods.	2	4	-	-
PC2. clean the parts for inspection.	2	4	-	-
PC3. check visual inspection on completion of all the process as per drawing.	3	4	-	-
PC4. check the calibration status of various measuring instrument.	3	3	-	-
PC5. identify parameter wise measuring instruments for measurement.	2	4	-	-
PC6. inspect the machined product with various measurement tools like vernier caliper, micrometer.	3	4	-	-
PC7. check the surface finish as per requirements.	2	3	-	-
PC8. check the fitness of the product using go or no-go devices.	3	4	-	-
PC9. check visual inspection on the damage, burr, excess materials.	2	3	-	-
PC10. document all the measured data for various parameters.	2	3	-	-
<i>Finalize and deliver the product</i>	16	24	-	-
PC11. clean the parts.	2	4	-	-
PC12. pack the parts in proper way to avoid any damages during transport.	2	3	-	-
PC13. complete the delivery documents.	2	3	-	-
PC14. deliver the parts, drawings, and digital memory devices to the appropriate location and/or personnel as required by the organization.	2	3	-	-
PC15. dismount tools, clamping, accessories.	2	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC16. clean the machines.	2	3	-	-
PC17. set the environment to its initial state, ready for the next job.	2	2	-	-
PC18. evaluate and report on factors and outcomes relevant to requirements and expectations.	2	3	-	-
NOS Total	40	60	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	CUTM/CSC/N0104
NOS Name	Inspect, Finalize, and deliver the product
Sector	Capital Goods
Sub-Sector	
Occupation	Additive Manufacturing
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2027
NSQC Clearance Date	31/01/2024

Qualification Pack

CUTM/VSQ/N7801: Ensure organizational health, safety, and security

Description

This NOS focuses on maintaining workplace safety, health, and security by identifying hazards, following safety procedures, using personal protective equipment (PPE), and responding to emergencies. It ensures compliance with health and safety regulations, proper waste disposal, and adherence to security protocols to create a safe and productive work environment.

Scope

The scope covers the following :

- This NOS covers workplace hazard identification, adherence to safety regulations, proper use of personal protective equipment (PPE), emergency response, fire safety, first aid, and waste management. It also includes implementing security measures, ensuring workplace hygiene, and promoting a safety-conscious culture to prevent accidents and maintain a secure working environment.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** identify different kinds of possible hazards at the workplace hazards: environmental, personal, ergonomic and chemical.
- PC2.** recognize other possible security issues existing in the workplace.
- PC3.** plan safety techniques for self and others.
- PC4.** recognize different measures to curb the hazards.
- PC5.** implement programs for health, safety and security of the people at the workplace.
- PC6.** communicate the safety plans to everyone at the workplace.
- PC7.** attach disciplinary rules with the implementation at the workplace.
- PC8.** comply with regulatory health and safety guidelines applicable to the workplace.
- PC9.** use and maintain protective equipment as per requirement in a production house.
- PC10.** ensure organizational activities in line with approved guidelines and procedures.
- PC11.** undertake and advise healthy lifestyle for self and others.
- PC12.** follow environment management system related procedures.
- PC13.** ensure safe handling and disposal of waste and debris.
- PC14.** minimize health and safety risks to self and others due to own actions.
- PC15.** monitor the workplace and work processes for potential risks and threats to self and others.
- PC16.** carry out periodic check to keep work area free from hazards and obstructions.
- PC17.** ensure participation of self and others in mock drills/ evacuation procedures organized at the workplace.
- PC18.** organize first aid, firefighting and emergency training sessions for others.
- PC19.** instruct and guide others to take action in the event of fire, emergencies or accidents.

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Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** safety guidelines, workplace policies, and compliance with national and industry safety standards.
- KU2.** common hazards such as electrical risks, fire, chemical exposure, and physical injuries.
- KU3.** different types of PPE (gloves, helmets, safety shoes, goggles) and their correct usage.
- KU4.** the proper handling, operation, and maintenance of workplace tools to prevent accidents.
- KU5.** Awareness of first aid procedures, fire evacuation plans, and emergency contact protocols.
- KU6.** fire hazards, fire extinguisher types, and safe fire-handling procedures in case of an incident.
- KU7.** the importance of keeping the workplace clean, organized, and free of potential hazards.
- KU8.** proper lifting posture, use of lifting tools, and avoiding strain-related injuries.
- KU9.** workplace security protocols, restricted areas, and unauthorized access prevention.
- KU10.** safe disposal methods for hazardous and non-hazardous materials to ensure environmental compliance.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** report safety concerns, follow instructions, and communicate hazards to supervisors and colleagues.
- GS2.** Identify workplace safety issues and taking appropriate actions to prevent accidents.
- GS3.** Handle safety-related measurements like voltage limits, weight capacity, and fire extinguisher ratings.
- GS4.** Ensure timely completion of tasks while following safety protocols and risk management procedures.
- GS5.** Work effectively with colleagues to maintain a safe and secure work environment.
- GS6.** Identify risks such as electrical shocks, fire, slippery surfaces, and chemical exposure in a workplace.
- GS7.** Ensure the safe use of machines, hand tools, and electrical equipment to prevent accidents.
- GS8.** Follow safety protocols, emergency procedures, and workplace safety policies.
- GS9.** do basic first aid techniques, fire safety procedures, and responding to workplace accidents.
- GS10.** Ensure proper disposal of hazardous and non-hazardous waste in compliance with environmental standards.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	70	-	-
PC1. identify different kinds of possible hazards at the workplace hazards: environmental, personal, ergonomic and chemical.	2	4	-	-
PC2. recognize other possible security issues existing in the workplace.	2	4	-	-
PC3. plan safety techniques for self and others.	2	4	-	-
PC4. recognize different measures to curb the hazards.	2	4	-	-
PC5. implement programs for health, safety and security of the people at the workplace.	2	4	-	-
PC6. communicate the safety plans to everyone at the workplace.	2	4	-	-
PC7. attach disciplinary rules with the implementation at the workplace.	2	4	-	-
PC8. comply with regulatory health and safety guidelines applicable to the workplace.	2	4	-	-
PC9. use and maintain protective equipment as per requirement in a production house.	2	3	-	-
PC10. ensure organizational activities in line with approved guidelines and procedures.	2	3	-	-
PC11. undertake and advise healthy lifestyle for self and others.	1	3	-	-
PC12. follow environment management system related procedures.	1	3	-	-
PC13. ensure safe handling and disposal of waste and debris.	2	4	-	-
PC14. minimize health and safety risks to self and others due to own actions.	1	4	-	-
PC15. monitor the workplace and work processes for potential risks and threats to self and others.	1	4	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC16. carry out periodic check to keep work area free from hazards and obstructions.	1	4	-	-
PC17. ensure participation of self and others in mock drills/ evacuation procedures organized at the workplace.	1	4	-	-
PC18. organize first aid, firefighting and emergency training sessions for others.	1	3	-	-
PC19. instruct and guide others to take action in the event of fire, emergencies or accidents.	1	3	-	-
NOS Total	30	70	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CUTM/VSQ/N7801
NOS Name	Ensure organizational health, safety, and security
Sector	Cross Sectoral
Sub-Sector	
Occupation	Health and Safety
NSQF Level	4
Credits	3
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings



Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

Assessment Strategy

1. Assessment System Overview:

- Batches would be assigned to Assessment Agencies as per the sectors for conducting assessment on CUTM portal or by email.
- Assessment agencies will have to give a confirmation regarding the assigned batches and share the particular Question set to Centurion University.
- Assessment agency deploys the ToA certified Assessor for conducting the assessment from the pool of verified assessor by Centurion University.
- Centurion University will monitor the assessment process and keep the records online or offline (wherever needed)

2. Procedure at Testing Centre:

- Check the Assessment location, date and time



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- Maximum batch size will be of 30.
- Assessor will Check that the allotted time to the candidates to complete Theory & Practical Assessment is correct.
- Assessor will ensure all the equipment's and required materials are available at the testing centre.

3. Assessment Quality Assurance:

- Question bank is created by the Subject Matter Experts (SME) of Assessment Agencies verified by Centurion University.
- Questions are mapped to the specified assessment criteria
- Assessor must be ToA certified & trainer must be ToT Certified
- Surprise visit would be done time to time.

4. Assessment Evidence collection:

- Geo tagged photographs of assessment sessions would be captured by the Assessor.
- Video call verification will be done by Centurion University during assessment and keep the recorded videos.

5. Document storing:

- Hard copies will be stored in the respective files.
- Some of documents will kept in both soft copies and hardcopies.
- All the assessment related data will be kept in CUTM portal.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Minimum Passing % at NOS Level: 70

(Please note: A Trainee must score the minimum percentage for each NOS separately as well as on the QP as a whole.)

Assessment Weightage

Compulsory NOS

Qualification Pack

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
CUTM/CSC/N0101.3D digitizing, scan to CAD and interpretation of designs	30	40	-	30	100	15
CUTM/CSC/N0102.Programming, Metrology, Maintenance, and repair of 3D machine	40	60	-	-	100	25
CUTM/CSC/N0103.Operate 3D Printing machine	40	60	-	-	100	20
CUTM/CSC/N0104.Inspect, Finalize, and deliver the product	40	60	-	-	100	20
CUTM/VSQ/N7801.Ensure organizational health, safety, and security	30	70	0	0	100	10
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
Total	200	320	0	0	550	100



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.